



Queen Adedire (Attempt

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Test Information

QUESTION 1: CALCULATED NUMERIC

10

out of 10 points

Calculate the coal consumption for the cycle. Express your answer in **kg/s** and round your numerical result into **a single decimal digit**.



Given Answer: 107.9

Correct Answer: 107.9

Answer range +/- 0.5 (107.4 - 108.4)

QUESTION 2: CALCULATED NUMERIC

10

out of 10 points

Calculate the temperature rise across the Feed Water Pump. Express your answer in **Kelvin** and round your numerical result into **a single decimal digit**.



Given Answer: 5.1

Correct Answer: 5.1

Answer range +/- 0.1 (5.0 - 5.2)

QUESTION 3: CALCULATED NUMERIC

10

out of 10 points

Calculate the thermal efficiency of the cycle. Express your answer in **%** and round your numerical result into **a single decimal digit**.



Given Answer: 38.3

QUESTION 4: CALCULATED NUMERIC

10

out of 10 points

Calculate the thermodynamic efficiency of the cycle. Express your answer in % and round your numerical result into **a single decimal digit**.



Given Answer: ✓ 50.6

Correct Answer: ✓ 50.6

Answer range +/- 0.5 (50.1 - 51.1)

QUESTION 5: CALCULATED NUMERIC

10

out of 10 points

Calculate the temperature of the stream **S8**. Express your answer in °C and round your numerical result into **a single decimal digit**.



Given Answer: ✓ 252.2

Correct Answer: ✓ 252.2

Answer range +/- 0.5 (251.7 - 252.7)

QUESTION 6: CALCULATED NUMERIC

10

out of 10 points

Calculate the amount of steam required for production of 750 MW of power from the two turbines. Express your answer in **kmol/s** and round your numerical result into **a single decimal digit**,



Given Answer: ✓ 34.4

Correct Answer: ✓ 34.4

Answer range +/- 0.5 (33.9 - 34.9)

QUESTION 7: CALCULATED NUMERIC

10

out of 10 points

Calculate the quality of the LP turbine steam discharge. Express your answer as fraction and round your numerical result into **three decimal digits**.



Given Answer: ✓ 0.977

Correct Answer: ✓ 0.977

Answer range +/- 0.002 (0.975 - 0.979)

QUESTION 8: CALCULATED NUMERIC**10****out of 10 points**

Calculate the amount of steam loss from the cycle. Express your answer in **KMTA** and round your numerical result into **a single decimal digit**.



KMTA: kilometric tonnes per annum [1 tonne = 1000 kg, 1 annum = 8000 hrs]

Given Answer: 35.9

Correct Answer: 35.8

Answer range +/- 0.5 (35.3 - 36.3)

QUESTION 9: CALCULATED NUMERIC**10****out of 10 points**

Calculate the lost work in the Open Feed Water Heater. Express your answer in **MW** and round your numerical result into **a single decimal digit**,



Given Answer: 35.4

Correct Answer: 35.4

Answer range +/- 0.5 (34.9 - 35.9)

QUESTION 10: CALCULATED NUMERIC**10****out of 10 points**

Calculate the steam flow to the Open Feed Water Heater. Express your answer in **kmol/s** and round your numerical result into **a single decimal digit**.



Given Answer: 6.9

Correct Answer: 6.9

Answer range +/- 0.2 (6.7 - 7.1)

FEEDBACK AND NOTES FOR ATTEMPT

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